

Argument-Based Framework for Quantitative Research

One quantitative question, two linked arguments, and three perspectives

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An empirical problem creates a need for an answer

Students enter quantitative-methods courses with very different preparation, and many struggle to connect statistical procedures with research reasoning.

This **empirical problem** matters, but it is not yet an answerable quantitative question. It does not identify whom the answer concerns, what will be observed, which relationship or comparison matters, or what kind of answer is sought.

A quantitative inquiry begins by converting the problem into one or more questions whose answers can be evaluated with empirical evidence.

FRAME Name the practical problem in one sentence, then ask what observable relationship would genuinely inform it.

Research check: A topic names an area; a problem identifies a consequential uncertainty; a research question specifies what must be answered.

Quantitative inquiry is one form of empirical research

Empirical research may be **quantitative, qualitative, or mixed-methods**. Quantitative inquiry is distinguished by questions answered through numerical description, estimation, comparison, association, or prediction.

Several classifications can coexist:

Perspective	Examples	What it clarifies
Intended claim	descriptive, associational, predictive, causal	the kind of answer sought
Design	observational, experimental, quasi-experimental	how evidence is produced
Time	cross-sectional, longitudinal	when and how often observations are made
Analytic status	confirmatory, exploratory	whether key decisions were specified prospectively

None of these labels alone determines whether an answer is credible. The question, evidence, support, analysis, and claim must work together.

CLASSIFY Choose one question and classify its intended claim, design, timing, and analytic status.

Research check: Labels can coexist; credibility depends on their alignment.

The same problem can generate different questions

Problem: students vary in preparation and later course performance.

Question type	Example quantitative question
Descriptive	What is the distribution of baseline quantitative-reasoning scores among entering students?
Associational	How is end-of-term performance related linearly to baseline score?
Predictive	How accurately can baseline information predict later performance for a new student?
Causal	What difference would assignment to retrieval practice make under a specified implementation?

Each question implies a different target, design burden, analysis, and bounded answer. A study may ask several questions, but it must not treat one analysis as an automatic answer to all of them.

COMPARE Explain why the descriptive, associational, predictive, and causal questions require different evidence.

Research check: A shared problem does not imply one interchangeable analysis.

Research question, hypothesis, and claim have different times

A **hypothesis** records a prospective, preferably prespecified candidate answer before the evidence is evaluated.

The hierarchy is simple:

- ▶ **Research question:** what the study seeks to answer.
- ▶ **Hypothesis:** an optional anticipated answer.
- ▶ **Claim:** the answer that remains defensible after the evidence is evaluated.

An answer has one of two **outcomes**: a bounded answer or an inconclusive answer under the available evidence. This is separate from **analytic status**, which is confirmatory or exploratory.

Purpose or objective may briefly explain why the study matters or what it will do; neither substitutes for a research question. **Target population** and **estimand** remain necessary because they have precise inferential meanings.

SEPARATE For one published study, identify its question, any prospective hypothesis, and its eventual claim.

Research check: A significant result is not itself a research question, and a hypothesis is not an evidence-based claim.

Two layers of answer require two linked arguments

For a substantive quantitative question:

1. the **statistical argument** develops an intermediate bounded claim about an estimand or predictive target; and
2. the **substantive argument** develops the final bounded answer in the question's disciplinary context.

A purely statistical, computational, or methodological question may stop with a bounded statistical claim. Component roles are therefore relative to the question being answered.

The arguments are logically parallel, and their eight components are functional roles rather than independent arguments or workflow stages:

$$(E + J) \longrightarrow I \longrightarrow C$$

- ▶ **Evidence** and **Justification** are co-premises. Evidence does not prove its own adequacy, and justification without evidence has nothing empirical to support.
- ▶ Together they support **Inference**; Inference supports a bounded **Claim**.
- ▶ A role may contain a subordinate argument. Score interpretation, missingness, or a meaningful benchmark can each require additional evidence and reasoning.

The components are logically ordered within each argument; the activities used to construct and revise them are operationally iterative.

FRAME Name the statistical answer and the final substantive answer before listing the eight roles.

Research check: The framework organizes support for one question; it is not an eight-step checklist.

Statistical Evidence and Justification

STAT-E - Empirical evidence

- ▶ Observed measurements and records, together with sampling or recruitment, assignment or exposure, measurement and timing, implementation, exclusions, missingness, and dependence information.
- ▶ It describes the study that actually occurred. Estimates, intervals, tests, R^2 , diagnostics, and sensitivity results are derived later and do not belong here.

STAT-J - Statistical justification

- ▶ Reasons the evidence, design, model, estimator, and uncertainty procedure can bear on the stated estimand or predictive target.
- ▶ It identifies necessary conditions, what can be checked, what must be argued, and what needs sensitivity analysis. It is not a ritual list declaring that assumptions “passed.”

DISTINGUISH Separate what was observed from why an inferential procedure is defensible.

Research check: Evidence and Justification are co-premises; neither can substitute for the other.

Statistical Inference and Claim

STAT-I - Statistical inference

- ▶ The reasoned analytical move from STAT-E under STAT-J to the declared target.
- ▶ It includes the estimand or predictive target, estimator or prediction rule, estimate or prediction, uncertainty or performance, checking, sensitivity, multiplicity, and analytic status.

STAT-C - Bounded statistical claim

- ▶ The intermediate answer about the estimand or predictive target.
- ▶ It preserves magnitude, direction, scale, uncertainty, model or design conditions, confirmatory or exploratory status, sensitivity, and unresolved statistical reservations.

A coefficient or significance label is not yet a bounded statistical claim.

BOUND Turn one numerical result into a complete statistical claim.

Research check: Keep the target, magnitude, uncertainty, conditions, and analytic status together.

Substantive Evidence and Justification

SUB-E - Contextualized empirical evidence

- ▶ What STAT-E represents substantively: the people or entities, measurements, conditions, setting, implementation, and period relevant to the question.
- ▶ It is not a second dataset or a relabelling exercise. It prevents an observed score, administrative category, or aggregate from acquiring an unsupported substantive meaning.

SUB-J - Substantive justification

- ▶ Why those contextualized observations and the statistical target bear on the research question.
- ▶ It draws on literature, theory, measurement meaning, design relevance, mechanisms and alternatives, meaningful benchmarks, and supported scope.

SITUATE Ask what the observed measures and represented units mean for the substantive question.

Research check: A disciplinary label does not by itself validate a construct or enlarge the population.

Substantive Inference and Claim

SUB-I - Substantive inference and interpretation

- ▶ Interprets STAT-I using SUB-E and SUB-J.
- ▶ It translates magnitude and uncertainty into substantive units, evaluates importance and alternatives, and identifies whether the supported meaning is descriptive, associational, predictive, or causal.

SUB-C - Bounded substantive claim

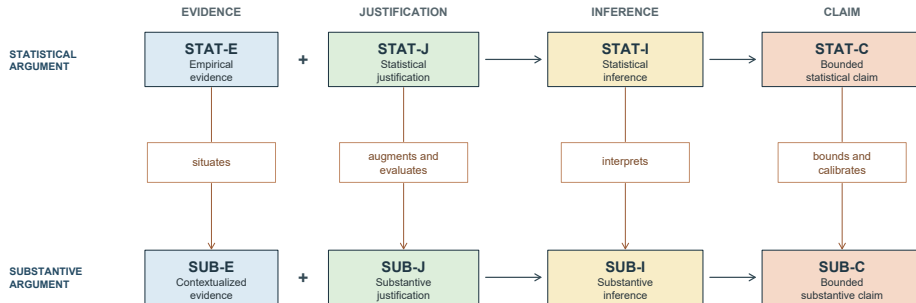
- ▶ The final answer to the research question.
- ▶ It states whom or what the answer concerns, setting and period, supported relationship or contrast, uncertainty, assumptions, alternatives, and limits on causal or population generalization.

It follows from SUB-I and cannot be broader or more certain than STAT-C.

INTERPRET Explain what the statistical result means in this context and why.

Research check: Substantive meaning may narrow a result; it cannot erase statistical uncertainty.

The complete structure is parallel and logically ordered



TRACE Read each argument from evidence and justification toward its claim, then compare matching roles vertically.

Research check: Parallel structure makes reasoning traceable; it does not imply that statistical and substantive reasoning perform identical operations.

Evidence transition: situates; Justification: augments and evaluates

STAT-E to SUB-E - situates

The observed record is tied to substantive referents: what the people or entities, variables, conditions, setting, and period mean. The transition must retain how evidence was produced. A test score does not become a latent construct, and an occupational average does not become an individual characteristic, merely by changing its label.

STAT-J to SUB-J - augments and evaluates

Design and statistical conditions remain visible while literature, theory, measurement evidence, substantive design relevance, alternatives, benchmarks, and scope are added and evaluated. A plausible theory cannot repair a weak comparison or unsupported uncertainty procedure.

TRACE For Evidence and Justification, identify what is retained and what substantive reasoning adds.

Research check: Situating evidence must preserve the actual unit, measurement, design, and statistical conditions.

Inference transition: interprets; Claim: bounds and calibrates

STAT-I to SUB-I - interprets

The estimate or prediction, uncertainty, checking, and sensitivity are expressed in substantive units and evaluated for meaning. Context can change whether a three-point difference matters; it cannot erase the interval around that estimate.

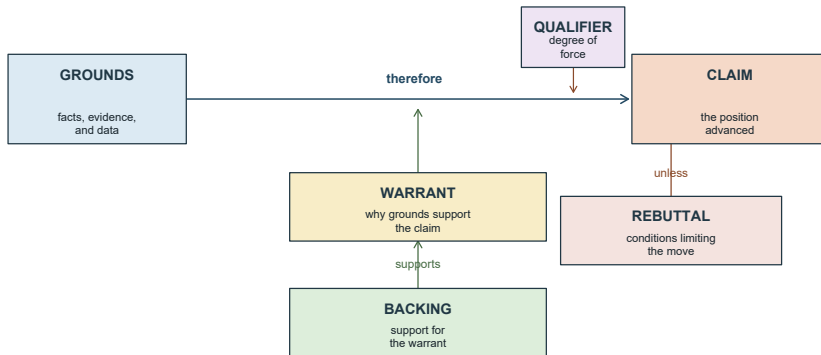
STAT-C to SUB-C - bounds and calibrates

SUB-C is developed through SUB-I, while STAT-C constrains how strong and broad the final answer may be. Statistical significance cannot be inflated into practical importance or causality.

CALIBRATE Carry the estimate, uncertainty, assumptions, and status into the substantive interpretation and claim.

Research check: Context supplies meaning, but the statistical claim still bounds the final answer.

Toulmin clarifies why evidence alone cannot carry a claim



TRACE Read from grounds through warrant to claim, then locate backing, qualifier, and rebuttal.

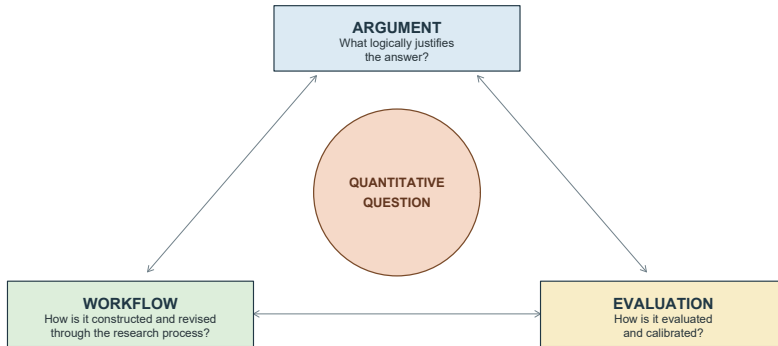
Research check: Evidence alone cannot determine the force or boundary of a claim.

Toulmin informs the framework without defining it

Toulmin's model highlights the inferential bridge, the force of a claim, and conditions that limit it (Toulmin, 2003). The present framework is **Toulmin-informed**, not a one-to-one recoding, and also develops LeMire's distinction between statistical and substantive argumentation beyond its original null-hypothesis-significance-testing setting (LeMire, 2010).

CONNECT Point to the grounds, warrant, qualifier, and reservation in one empirical claim.
Research check: The framework is Toulmin-informed, not a one-to-one relabelling.

Three perspectives keep the quantitative question at the centre



ORIENT Use all three perspectives on the same question.

Research check: Each perspective exposes a different part of the reasoning without creating a new research question.

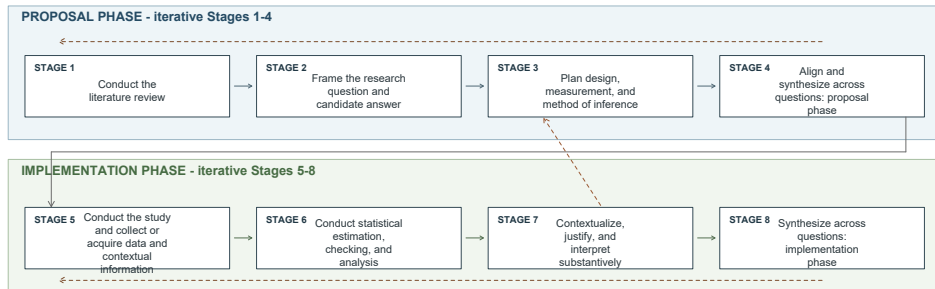
Each perspective has one distinct job

- ▶ **Argument:** What must logically justify the answer?
- ▶ **Workflow:** How is the argument constructed and revised through the research process?
- ▶ **Evaluation:** How is the argument evaluated and calibrated?

ORIENT Ask the same question through Argument, Workflow, and Evaluation.

Research check: The perspectives coordinate different views of one answer.

The workflow is iterative, not a straight checklist



Stages are activities, not structural components. One stage may develop several components, and a later finding may require a disciplined return to an earlier decision.

REVISE Identify a result that would send the inquiry back to an earlier stage.

Research check: Iteration should be motivated and documented, not hidden.

Eight workflow stages span proposal and implementation

Proposal phase	Implementation phase
1. Conduct the literature review.	5. Conduct the study and collect or acquire data and contextual information.
2. Frame the research question and candidate answer.	6. Conduct statistical estimation, checking, and analysis.
3. Plan design, measurement, and method of inference.	7. Contextualize, justify, and interpret substantively.
4. Align and synthesize across questions.	8. Synthesize across questions.

Proposal anticipates evidence and support without treating a candidate answer as established. Implementation gives every question an answer while preserving reservations. Later evidence may transparently reopen an earlier decision.

REVISE Locate the stage to revisit when evidence challenges an earlier decision.

Research check: Operational iteration is legitimate when the reason and consequence are visible.

Seven evaluation lenses locate vulnerable reasoning

1. **Question alignment** – question, target, design, method, claim type.
2. **Evidence integrity** – production, measurement, missingness, dependence.
3. **Justification adequacy** – statistical and substantive support.
4. **Inferential adequacy** – target, method, checking, uncertainty, status.
5. **Transition fidelity** – preservation across all four corresponding roles.
6. **Claim calibration** – strength and scope no greater than support permits.
7. **Cross-question coherence** – distinct answers and transparent synthesis.

The lenses are many-to-many, not a total score or a one-to-one component map.

EVALUATE Identify the lens that most constrains the claim rather than counting strengths.

Research check: One decisive weakness can outweigh several favorable features.

Lead example: context, question, hypothesis, and method

An educational team prospectively follows 310 first-year students who complete one quantitative-methods course. Students complete a 0-100 quantitative-reasoning assessment in the first week and a 0-100 end-of-term assessment after the common twelve-week course. No study condition is assigned, and all 310 course completers have both scores.

Research question

Among first-year students completing the course, how is end-of-term performance related linearly to baseline quantitative-reasoning score?

Prospective hypothesis: the working-population slope is positive.

The prespecified model is

$$E(Y \mid X) = \beta_0 + \beta_1(X - 50).$$

Suppose $\hat{\beta}_1 = 0.42$ with a 95% confidence interval from 0.30 to 0.54. The detailed conditions for this interval are introduced in Section 4; here the result shows how uncertainty must remain visible in both arguments.

SPECIFY Underline the population, unit, variables, design, model, and hypothesis.

Research check: The estimate is interpretable only after its target and evidence are defined.

Lead example: statistical evidence and justification

Component	Application
STAT-E	Baseline and end-of-term scores for 310 course completers, with recruitment, assessment timing, scoring, completion, and study-period information.
STAT-J	The prespecified working model, consistent assessment, complete outcomes for the analyzed cases, and appropriately independent student information support the slope interval over the represented range.

TRACE Explain how the evidence and justification support the declared slope.

Research check: The data record and the model conditions perform different roles.

Lead example: statistical inference and claim

Component	Application
STAT-I	Confirmatory slope estimate 0.42 points per baseline point; 95% CI [0.30, 0.54].
STAT-C	For the defined working population and model, the slope is estimated as 0.42, 95% CI [0.30, 0.54].

TRACE Follow the slope from STAT-E and STAT-J through STAT-I to STAT-C.

Research check: The intermediate claim retains the target, scale, uncertainty, and model conditions.

Lead example: substantive evidence and justification

Component	Application
SUB-E	The evidence concerns students completing one course and two assessments of quantitative performance in one institution and intake.
SUB-J	The assessments support a bounded achievement interpretation; completion selection and the observational design limit generalization and causal interpretation.

TRACE Explain what substantive context adds while retaining the statistical evidence and conditions.

Research check: Meaning does not enlarge the represented population or change the design.

Lead example: substantive inference and claim

Component	Application
SUB-I	A 10-point baseline difference corresponds to an estimated 4.2-point end-of-term difference, 95% CI [3.0, 5.4].
SUB-C	Among represented course completers, a 10-point higher baseline score was associated with an estimated 4.2-point higher expected score, 95% CI [3.0, 5.4]; the evidence does not show what changing preparation would cause.

Answer outcome: bounded positive association. **Analytic status:** confirmatory.

TRACE Follow SUB-E and SUB-J through SUB-I to the final answer in SUB-C.

Research check: The substantive argument adds meaning while retaining magnitude, uncertainty, and limits.

Lead example: proposal workflow

Stage	Concrete application
1. Literature review	Examine entry preparation, course learning, assessment meaning, and plausible nonlinear relationships.
2. Question and candidate answer	State the associational question and prospectively record the positive-slope hypothesis.
3. Design, measurement, and inference plan	Define course completers, the student unit, two assessments, slope target, model, interval, and checks.
4. Proposal synthesis	Check that population, measurements, estimand, and noncausal interpretation align.

During piloting, a ceiling in the baseline assessment would return the team to Stages 1-3 before the focal study.

LOCATE At which stage would a ceiling effect force revision?

Research check: A measurement problem should reopen planning before the focal analysis.

Lead example: implementation workflow

Stage	Concrete application
5. Study and evidence production	Administer both assessments and document participation, completion, timing, scoring, and instruction.
6. Statistical analysis	Estimate and check the slope and interval over the represented score range.
7. Substantive interpretation	Translate to a 10-point contrast and retain the observational and completer-only boundaries.
8. Implementation synthesis	State the bounded answer and keep prediction, subgroup, and intervention questions separate.

Iteration strengthens alignment when its reason is visible; it is not a post-result search for a favorable coefficient.

REVISE Name a result that should reopen an earlier planning decision.

Research check: Transparent revision changes the argument and its analytic status when needed.

Lead example: Evaluation perspective - alignment and inference

Lens	Evaluation
Question alignment	The slope, working population, assessments, and associational wording agree.
Evidence integrity	Timing, scoring, and complete outcomes are documented; selection into course completion still determines representation.
Justification adequacy	The model and observational design can support association, not an intervention effect or transport to another institution.
Inferential adequacy	The interval depends on the linear summary, represented range, independence, measurement, and uncertainty procedure.

DIAGNOSE Which of these four lenses most constrains the result?

Research check: A useful evaluation identifies the weakness that changes the answer

Lead example: Evaluation perspective - transition and synthesis

Lens	Evaluation
Transition fidelity	The 10-point interpretation retains magnitude, interval, confirmatory status, and completer-only scope.
Claim calibration	“Higher baseline scores were associated with higher performance” is supportable; “raising preparation will improve performance” is not.
Cross-question coherence	Prediction, mechanism, subgroup, and causal questions require distinct targets and arguments.

CALIBRATE Which tempting interpretation is ruled out by the design?

Research check: The bounded answer is associational and applies to represented course completers.

Several questions require several linked arguments

Questions may share participants, measurements, literature, and even one fitted model. They still require separate arguments when they have different estimands, comparisons, assumptions, or intended interpretations.

For example:

- ▶ **RQ1:** How is later performance related to baseline preparation?
- ▶ **RQ2:** Does that association differ between two instructional settings?

The second question targets an interaction, not merely another coefficient in the first argument. Each question needs its own statistical claim, substantive claim, and answer outcome. The study then synthesizes the linked answers.

SEPARATE Give each question its own estimand and bounded answer.

Research check: Shared evidence does not make different questions one argument.

Population, observational unit, and setting bound the answer

A research question should remain readable, while its research specification identifies:

- ▶ the **target population** to which the answer is intended to apply;
- ▶ the **observational unit** represented by each observation;
- ▶ the outcome, comparison, setting, and time;
- ▶ the estimand or predictive target; and
- ▶ the intended descriptive, associational, predictive, or causal claim type.

These decisions shape **STAT-E**, **STAT-I**, and the scope of both claims. A sample can contain students while the target is schools, or occupations while the intended interpretation concerns individuals; such mismatches are not repaired by a more elaborate model.

BOUND State the population, unit, setting, and period for one proposed claim.

Research check: A claim cannot be broader than the evidence-producing design.

The estimand states what the statistical answer is about

An **estimand** is the population or model quantity the question asks us to learn. An **estimate** is the value calculated from observed evidence.

Examples include:

- ▶ a population mean or proportion;
- ▶ a mean difference between specified groups;
- ▶ a regression slope under a stated conditional-mean model;
- ▶ a predicted probability for a defined population and setting; or
- ▶ a difference in prediction error under a defined deployment task.

Without a target, an estimate has no stable interpretation. The estimator and model belong to the method of learning about the target, not to the target itself.

TARGET Rewrite one broad question as a precise population quantity or prediction target.

Research check: The estimand fixes what the estimate is intended to approximate.

Association, prediction, and causation require different support

Intended claim	Central question	Additional burden
Description	What pattern is present?	measurement and scope
Association	How do variables vary together?	conditional meaning and alternatives
Prediction	How well will outcomes be predicted in a specified setting?	validation matched to deployment
Causation	What would changing an exposure do?	identification through design and assumptions

Regression can estimate all four kinds of target, but the regression equation does not determine which claim type is justified. The design and substantive reasoning do that work.

MATCH Classify a claim as descriptive, associational, predictive, or causal and name the support it needs.

Research check: A model form does not determine the claim type.

Observable and latent variables require different frameworks

An **observable variable** records a directly observed or coded quantity, such as age, course mark, treatment assignment, or occupational average income.

A **latent variable** represents a construct that is not directly observed, such as belonging, anxiety, or academic self-efficacy. It usually requires multiple indicators and an explicit measurement model.

This regression-based approach mainly treats observed variables as defined. When the research question concerns the adequacy of a construct or its measurement, measurement evidence becomes the object of its own argument. Factor analysis and latent-variable modelling therefore require a related but distinct framework.

DISTINGUISH Identify what is directly observed and what remains a construct interpretation.

Research check: A score is an observable measure, not automatically the latent attribute.

Missingness changes both evidence and meaning

Missing outcomes or predictors can change:

- ▶ which population the analysis represents;
- ▶ the estimand that is effectively being estimated;
- ▶ the assumptions needed for inference; and
- ▶ the strength of the substantive answer.

Missing at random is an untestable conditional assumption, not a property confirmed from observed values alone. A method relying on it must state the conditioning information and pair the assumption with appropriate sensitivity analysis.

BOUND Ask what population remains represented after participation, follow-up, and missingness.

Research check: A complete analysis of incomplete evidence requires both a method and a defensible account of the process that produced missingness.

Sampling and assignment solve different problems

target population $\supseteq$ sampling frame $\supseteq$ invited sample $\supseteq$ respondents.

- ▶ **Probability sampling** can support population generalization when its implementation and nonresponse are adequately addressed.
- ▶ **Random assignment** can support a causal comparison within the assigned study population when implementation, follow-up, and interference are adequately addressed.

Random assignment does not automatically make a convenience sample representative. Representative sampling does not automatically remove confounding from an observational exposure.

SEPARATE Ask whether chance selected units, assigned conditions, both, or neither.

Research check: Sampling supports population extension; assignment supports treatment comparison.

Observational, experimental, and quasi-experimental designs

Observational design: exposure is not assigned by the researcher. It can support description, association, and prediction; causal interpretation needs additional identification arguments and remains vulnerable to alternatives.

Randomized experimental design: assignment is governed by chance. It can support a causal comparison of assigned conditions when implementation, outcome observation, interference, and analysis preserve the randomization logic. Researcher control without random assignment does not, by itself, provide the same causal support.

Quasi-experimental design: assignment follows a rule, threshold, timing, or external event rather than randomization. Credibility depends on design-specific assumptions and checks.

Design structures such as blocking, repeated observations, clustering, and factorial assignment modify these broad families rather than create a separate claim type.

COMPARE For each design family, name its strongest supported claim and central reservation.

Research check: Regression encodes a comparison but cannot supply the design.

Analysis and modelling support two forms of inference

Activity or component	What it does	What it cannot supply by itself
Statistical analysis and modelling	describes, formalizes, estimates or predicts, checks, and probes sensitivity	a justified target, reference process, or substantive meaning
Statistical inference (STAT-I)	uses STAT-E and STAT-J to infer about an estimand or predictive target; supports STAT-C	practical importance or broader substantive scope
Substantive inference (SUB-I)	uses SUB-E and SUB-J to interpret STAT-I; supports SUB-C	permission to discard uncertainty or design limits

The two inference components support but remain distinct from their claims. SUB-I may support descriptive, associational, predictive, or causal meaning. Causal inference is neither a third layer nor a synonym for substantive inference: the statistical argument must define, identify, and estimate a causal target, while the substantive argument establishes its meaning and scope.

SEPARATE Distinguish producing a model result from justifying statistical and substantive inferences.

Research check: 'STAT-I' and 'SUB-I' are inferential roles; 'STAT-C' and 'SUB-C' are the claims they support.

Inferential conditions belong to the argument

Different methods need different conditions, but recurring families include:

- ▶ **target and specification:** population, estimand or predictive target, variables, contrast, scale, and functional form;
- ▶ **evidence production:** sampling, recruitment, assignment, measurement, timing, implementation, and coverage;
- ▶ **stochastic reference process:** the source of variation represented by an interval, test, or prediction assessment;
- ▶ **dependence and interference:** clusters, repeated observations, temporal relation, and spillovers;
- ▶ **selection and missingness:** participation, attrition, exclusions, assumption-compatible methods, and sensitivity;
- ▶ **analytic conduct:** prespecification, exploration, multiplicity, model selection, checking, and sensitivity.

Diagnostics can reveal incompatibility or fragility. They do not prove a model true or make an untestable assumption disappear.

CHECK Name the inferential condition that gives an interval, test, or prediction assessment its interpretation.

Research check: Diagnostics can expose fragility; they cannot prove assumptions or identification conditions true.

Time and dependence determine how much information is present

Repeated observations from one student, students in one class, or occupations within one sector may be dependent. Treating correlated observations as independent can make uncertainty look too small.

Time also changes the question:

- ▶ cross-sectional association concerns one period;
- ▶ longitudinal change concerns within-unit development;
- ▶ lagged prediction concerns a specified future horizon; and
- ▶ intervention follow-up concerns an outcome after assignment.

The model, uncertainty procedure, and substantive scope must respect the design's actual information structure.

CHECK Identify the independently informative unit and the time ordering.

Research check: Repeated or clustered observations do not create independent information.

Research documentation supports traceability

Document enough to reconstruct:

- ▶ the problem, question, hypothesis where present, and estimand or predictive target;
- ▶ population, unit, level, eligibility, recruitment, sampling, and assignment or exposure;
- ▶ instruments, scoring, measurement timing, collection or acquisition, and intervention delivery;
- ▶ inclusion, exclusion, missingness, dependence, interference, and deviations from the plan; and
- ▶ the model, estimator, uncertainty procedure, checking, sensitivity, and analytic status.

Documentation is not clerical decoration. It helps establish STAT-E and STAT-J, gives substantive meaning to SUB-E, and explains the boundaries of both claims.

TRACE Name the documentation needed to reconstruct the intended comparison and implemented study.

Research check: Traceability links the question and plan to what actually occurred.

Hypothetical example: randomized retrieval practice

One university is deciding whether to replace standard study support with guided retrieval practice. Prior research suggests that active recall can strengthen delayed access to learned material. The university has identified a two-point difference on its delayed assessment as potentially important.

Research question: among the 240 consenting first-year students in this intake, what difference in mean delayed performance would result from assignment to retrieval practice rather than standard review in this four-session implementation?

Prespecified hypothesis: the mean assignment effect for these students, standardized over their baseline-score distribution, is at least the two-point importance benchmark.

FRAME Separate the practical decision, research question, and prespecified hypothesis.

Research check: The importance benchmark belongs to the substantive argument, not the p-value.

Randomized study: population, design, and measurement

Feature	Specification
Population and unit	240 consenting first-year students in one university intake; the student is the assignment, measurement, and analysis unit.
Design and implementation	Individual random assignment, 120 per condition; four sessions matched in duration and content coverage; delivery and attendance recorded.
Measurement and timing	A 0-40 baseline assessment before assignment and a parallel 0-40 delayed assessment two weeks after the fourth session; common scoring and complete follow-up.

AUDIT Check that the population, assignment, implementation, and measurement describe one coherent study.

Research check: Random assignment must remain connected to the conditions and outcome actually observed.

Randomized study: target, status, and scope

Feature	Specification
Target and status	Marginal mean assigned-group contrast for Retrieval rather than Standard, standardized over the empirical baseline-score distribution of the 240 consenters; a prespecified additive model estimates this target, randomization supports its assignment-effect interpretation, and comparisons against zero and two points are confirmatory.
Scope	Randomization can support causality for assignment for these students and this implementation under consistency, measurement, follow-up, and interference conditions; consent and one setting do not support wider generalization without additional evidence.

The primary comparison follows assignment. It is not automatically the effect of receiving every session under nonadherence.

DISTINGUISH Separate the marginal assignment estimand from its adjusted estimator.

Research check: The additive model estimates the declared target only because its assignment contrast is constant across baseline values.

Randomized study: one marginal target

Let $Z = 1$ denote Retrieval assignment and $Z = 0$ Standard assignment, let X_0 be the baseline score, and define $m_z(x) = E(Y \mid Z = z, X_0 = x)$. If G is the empirical baseline distribution of the 240 consenters, the target is

$$\Delta_G = E_{X_0 \sim G}[m_1(X_0) - m_0(X_0)].$$

Both assigned formats are standardized to the same G , so the estimand answers the marginal research question for these students.

TARGET Identify the contrast and the distribution over which it is averaged.

Research check: A standardized contrast is not fully defined until the target baseline distribution is named.

Randomized study: why the adjusted coefficient targets it

The prespecified baseline-adjusted model is

$$E(Y \mid Z = z, X_0 = x) = \beta_0 + \beta_1 z + \beta_2(x - 20).$$

With no assignment-by-baseline interaction, the conditional contrast is the constant β_1 at every baseline value. Averaging both assigned formats over the same G gives $\Delta_G = \beta_1$. Random assignment, well-defined support formats, complete follow-up, consistency, and no material interference support interpreting this assigned-group contrast as an effect of assignment for these students.

ALIGN Explain why β_1 answers the marginal question.

Research check: Additivity makes the conditional contrast constant; common-distribution standardization then leaves it unchanged.

Randomized study: conditions and result

Suppose $\widehat{\Delta}_G = 3.1$ points, 95% CI [0.8, 5.4]. The interval is positive but includes values below and above the two-point benchmark.

- ▶ The assignment interpretation uses randomization, consistency, stable measurement, complete follow-up, and no material interference.
- ▶ Exact classical t inference additionally uses a full-rank additive model with independent, conditionally Gaussian constant-variance errors.
- ▶ Confirmatory status depends on preserving the prespecified question, estimand, model, outcome, and benchmark.

INTERPRET Compare the interval with both zero and the two-point benchmark.

Research check: Evidence for a positive effect is not proof that it exceeds the importance threshold.

Statistical argument: evidence and justification

Component	Contribution
STAT-E	Assignment, baseline and delayed scores, scoring, session delivery, attendance, consent, and complete follow-up for 240 students.
STAT-J	Random assignment supports the assigned-group comparison; the no-interaction additive model makes β_1 equal the marginal contrast standardized to the same baseline distribution in both groups. Complete outcomes and the stated error conditions support the interval.

TRACE Identify how the evidence and justification support the declared estimand.

Research check: Randomization and the model play different roles in the statistical argument.

Statistical argument: inference and claim

Component	Contribution
STAT-I	Estimated baseline-standardized marginal assignment contrast 3.1 points; 95% CI [0.8, 5.4]; the interval exceeds zero and includes effects around the two-point benchmark.
STAT-C	For the empirical baseline distribution of these 240 students, the prespecified confirmatory analysis estimates the marginal assignment contrast as 3.1 points, 95% CI [0.8, 5.4]; values below and above the benchmark remain compatible with the evidence.

TRACE Check that the bounded statistical claim names the estimand, target distribution, interval, benchmark, and status.

Research check: A coefficient becomes an answer only after its target and uncertainty are explicit.

Substantive argument: evidence and justification

Component	Contribution
SUB-E	Consenting first-year students were assigned to four matched study-support sessions and completed a 0-40 delayed assessment in one intake.
SUB-J	Retrieval-practice theory, documented delivery, and randomization support an assignment-based educational interpretation; the evidence does not identify receipt effects under nonadherence.

CONNECT Relate the study record and contextual support to their statistical counterparts.

Research check: Substantive interpretation retains the assigned-format contrast and its implementation boundaries.

Substantive argument: inference and claim

Component	Contribution
SUB-I	For these 240 consenters' baseline distribution, the estimated mean assignment effect is 3.1 points, 95% CI [0.8, 5.4]; the evidence does not establish that the effect exceeds the benchmark.
SUB-C	Under the stated consistency, measurement, follow-up, and interference conditions, Retrieval rather than Standard assignment is estimated to raise mean delayed performance for these 240 students by 3.1 points, 95% CI [0.8, 5.4]. An effect above two points and extension beyond this intake and implementation remain uncertain.

Answer outcome: bounded positive causal answer about assignment for these students in this implementation. **Analytic status:** confirmatory.

CONNECT Relate each substantive component to its statistical counterpart.

Research check: The final claim retains assignment, magnitude, interval, benchmark, and scope.

Argument perspective: trace all four relations

- ▶ **Evidence - situates:** assignment and scores become evidence about four study-support sessions and delayed achievement for consenting students in one intake.
- ▶ **Justification - augments and evaluates:** randomization and model conditions are considered together with retrieval-practice theory, implementation fidelity, assessment meaning, and population scope.
- ▶ **Inference - interprets:** the baseline-standardized 3.1-point effect and its interval are interpreted on the 0-40 scale and compared with the two-point benchmark.
- ▶ **Claim - bounds and calibrates:** a positive assignment effect is supportable for these students in this implementation; an effect certainly above two points or an extension to other students or settings is not.

Randomization strengthens causal interpretation within the study. It does not settle measurement, implementation, generalization, or practical importance.

EXPLAIN For each relation, say what is retained and what context adds.

Research check: Randomization strengthens one relation; it does not settle every relation.

Workflow perspective: randomized-study proposal

Stage and activity	Concrete decision or result
1. Conduct the literature review	Review retrieval-practice theory and evidence about delayed learning.
2. Frame the research question and candidate answer	Frame the marginal assignment question for the 240 consenters and prespecify an effect of at least two points.
3. Plan design, measurement, and method of inference	Define the target baseline distribution and marginal estimand; plan random assignment, matched delivery, the additive no-interaction model, interval, checks, and benchmark.
4. Align and synthesize across questions: proposal phase	Verify that question, design, target distribution, estimand, coefficient-to-estimand link, and benchmark address the same comparison.

ALIGN Explain why baseline adjustment belongs at Stage 3 rather than inside the substantive question.

Research check: The question names the marginal assignment effect; the plan explains when the adjusted coefficient estimates it.

Workflow: randomized-study implementation

Stage and activity	Concrete decision or result
5. Conduct the study and collect or acquire data and contextual information	Recruit 240 consenters, assign 120 per condition, deliver four sessions, and record complete outcomes.
6. Conduct statistical estimation, checking, and analysis	Fit and check the additive model; estimate the baseline-standardized marginal effect as 3.1 points, 95% CI [0.8, 5.4].
7. Contextualize, justify, and interpret substantively	Interpret the estimate using randomization, implementation evidence, theory, and the importance benchmark.
8. Synthesize across questions: implementation phase	State a bounded positive answer, report confirmatory status separately, and preserve the limit on wider generalization.

If delivery or follow-up differs by condition, Stage 5 sends the inquiry back to the design assumptions and planned interpretation rather than simply adding another coefficient.

REVISE Choose one implementation problem and identify which earlier planning stage it reopens.

Research check: A deviation changes the argument; it is not repaired by an unplanned coefficient alone.

Evaluation perspective: alignment, evidence, and inference

Lens	Evaluation
Question alignment	The question, assignment contrast, outcome timing, baseline-standardized marginal target, and intended causal scope agree.
Evidence integrity	Documented delivery and complete outcomes are strengths; interference and assessment consistency still require scrutiny.
Justification adequacy	Randomization, consistent implementation, no material interference, and retrieval-practice theory support the two moves, while one setting limits scope.
Inferential adequacy	The coefficient equals the declared marginal effect only under the additive no-interaction model and common target baseline distribution; baseline nonlinearity and assignment-by-baseline variation deserve sensitivity checks.

EVALUATE Which of these four lenses most threatens the causal assignment interpretation?

Research check: A model check cannot substitute for randomization, implementation, or measurement support.

Evaluation: transition, calibration, and coherence

Lens	Evaluation
Transition fidelity	The target baseline distribution, 3.1-point estimate, interval, analytic status, benchmark, assigned-format meaning, and study-population boundary remain visible.
Claim calibration	A causal assignment claim is supportable for these 240 consenters in this implementation; extension beyond them or an effect certainly above two points is not established.
Cross-question coherence	With one focal question, coherence means consistency among theory, design, estimand, analysis, and answer; any mechanism question needs its own argument.

Revision rule: a serious model or implementation problem weakens the claim or triggers a transparent sensitivity analysis; it does not justify a retrospectively stronger story.

EVALUATE Find the lens that most narrows the randomized-study claim.

Research check: Evaluation should change the wording or trigger a transparent revision.

What the randomized application integrates

- ▶ **Section 2:** two argument structures, four role-specific transitions, an eight-stage Workflow, and seven Evaluation lenses.
- ▶ **Section 3:** student population and unit, observable assessments, a baseline-standardized marginal assignment estimand, evidence production, timing, and complete follow-up.
- ▶ **Section 4:** random assignment as support for the assignment comparison, model conditions for the interval, implementation and interference, and the different support needed for causality and generalization.

Regression supplies the standardized marginal comparison and uncertainty under the additive model. Design, measurement, theory, implementation, and context determine what that result can mean for the research question.

SYNTHESIZE Explain how Sections 2–4 jointly support the randomized answer.

Research check: Regression supplies the comparison; design and context determine its meaning and scope.

Real example: context and empirical evidence

The public **Prestige** data support a historical question about how education, economic rewards, and social standing varied across Canadian occupations. They describe 102 occupations using information associated with the 1971 Census and an occupational-prestige study (Fox et al., 2022; Pineo & Porter, 1967).

- ▶ The unit is an occupation, not an individual worker.
- ▶ Income is average occupational income in 1971 Canadian dollars (CAD).
- ▶ Education is average years of education; prestige is an occupational rating; women is workforce percentage; and type is a broad occupational group.
- ▶ The design is observational, cross-sectional, historical, and aggregate; the occupations are not identified as a probability sample.

Both analyses use the same 98 occupations with complete information for all variables used in either model. Occupational type is unavailable for four of the original 102 occupations, so these 98 form the empirical basis. The targets are working-superpopulation projections for comparable occupations; wider extension is model-based and conditional rather than design-based.

BOUND State what one observation represents and why the evidence is aggregate and historical.

Research check: Occupation-level evidence cannot support an individual-worker claim.

Real example: questions, hypothesis status, and targets

RQ1 - marginal: Among the represented occupations, how is average income related linearly to average education?

RQ2 - conditional: Among the same occupations, how is income related linearly to education at the same modelled prestige, percentage of women, and occupational type?

Hypothesis status: no prospective hypotheses are claimed. Both are exploratory reanalyses of historical evidence.

These are linked questions with different estimands—not competing answers to one question.

Question	Method and target
RQ1	Simple linear model; marginal occupation-level education slope.
RQ2	Additive multiple regression; education slope at the same modelled prestige, percentage of women, and occupational type.

SEPARATE Underline how RQ2 changes the estimand rather than merely adding predictors.
Research check: Marginal and conditional coefficients answer different questions.

Real example: inferential conditions

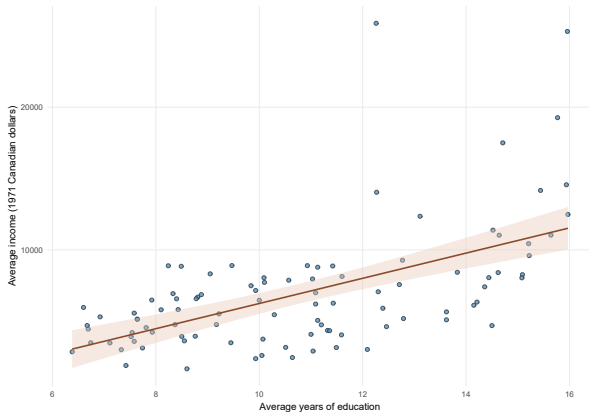
Each slope is a working-superpopulation projection for comparable occupations under the historical measurement regime.

- ▶ Exact classical t intervals require the corresponding full-rank linear conditional-mean model, independent occupations, and conditionally Gaussian errors of constant variance.
- ▶ Functional form, influence, and overlap in RQ2 require checking; otherwise the intervals are model-based approximations whose fragility enters the claims.
- ▶ Restricting both questions to the same 98 occupations separates model change from sample change, but wider extension requires additional assumptions.
- ▶ The intervals are not design-based sampling intervals and do not quantify uncertainty in the historical measurements.
- ▶ Both analyses remain exploratory.

CHECK Name which conditions support the interval and which limit generalization.

Research check: A model-based interval does not become design-based sampling evidence.

The marginal pattern is positive at the occupation level



The figure is evidence about occupations, not individual workers. Its slope cannot be read as an individual's return to another year of education.

READ Describe direction, form, scatter, support, and unusual occupations before reading the slope.

Research check: The fitted line summarizes an occupation-level pattern, not an individual return.

RQ1: statistical argument for the marginal association

Component	Contribution
STAT-E	The same 98 occupations with average income and education.
STAT-J	A working-superpopulation linear mean model, independent occupations, and conditionally Gaussian constant-variance errors support the classical interval; form and influence require checking.
STAT-I	Education slope CAD 883; 95% CI [CAD 628, CAD 1,138]; exploratory analysis.
STAT-C	In the exploratory marginal model, the slope is CAD 883, 95% CI [CAD 628, CAD 1,138].

TRACE Carry the target, model conditions, slope, interval, and exploratory status into STAT-C.

Research check: A numerical result becomes a bounded statistical claim only when its inferential conditions remain visible.

RQ1: substantive argument for the marginal association

Component	Contribution
SUB-E	The measures are occupation-level summaries from the represented 1971 period; income is in Canadian dollars.
SUB-J	Occupational sorting and credential requirements make association plausible; aggregation and observational design limit interpretation.
SUB-I	One additional occupation-average education year corresponds to CAD 883 higher average income, 95% CI [CAD 628, CAD 1,138], in 1971 Canadian dollars.
SUB-C	For comparable occupations under the historical working model, one additional average education year was associated with CAD 883 higher average income, 95% CI [CAD 628, CAD 1,138]; this is neither an individual return nor a causal effect.

Answer outcome: bounded positive occupational association. **Analytic status:** exploratory.

INTERPRET Explain why the same slope cannot be read as an individual return to schooling.

Research check: Units, historical context, and observational design bound the substantive claim.

RQ2: statistical argument for the conditional association

Component	Contribution
STAT-E	The same 98 occupations, adding prestige, percentage of women, and occupational type.
STAT-J	The additive model, adequate overlap, independent occupations, and conditionally Gaussian constant-variance errors support the classical interval; form and influence require checking.
STAT-I	Conditional education slope CAD 116; 95% CI [CAD -430, CAD 663]; exploratory analysis.
STAT-C	In the exploratory additive model, the conditional slope is CAD 116, 95% CI [CAD -430, CAD 663].

COMPARE Identify what changed from the marginal to the conditional estimand.

Research check: Conditioning changes the comparison even though the represented occupations remain the same.

RQ2: substantive argument for the conditional association

Component	Contribution
SUB-E	All measures remain occupation-level summaries in the represented 1971 period.
SUB-J	Comparing occupations at equal modelled prestige, workforce composition, and type asks whether the education-income pattern remains within broad occupational contexts; these variables are not automatically causal controls.
SUB-I	Per additional occupation-average education year, the interval is compatible with negative, negligible, or moderately positive income differences in 1971 Canadian dollars.
SUB-C	For comparable occupations under the historical working model and at the stated covariate values, the association is CAD 116, 95% CI [CAD -430, CAD 663], per additional occupation-average education year; the answer is inconclusive and model-dependent.

Answer outcome: inconclusive. **Analytic status:** exploratory.

INTERPRET Explain why an interval spanning negative and positive values makes the conditional answer inconclusive.

Research check: Inconclusive is not evidence of no relationship.

Cross-question synthesis explains why both results can be true

Question	Estimand	Result	Bounded substantive answer
RQ1 marginal	occupation-level linear slope without conditioning	CAD 883 [CAD 628, CAD 1,138]	positive occupation-level association
RQ2 conditional	education slope at equal modelled prestige, percentage of women, and type	CAD 116 [CAD -430, CAD 663]	inconclusive conditional association

The coefficient change is not caused by a change in the represented occupations. It arises because the two models answer different questions.

The synthesis does not ask which coefficient is the “real effect.” It states what each estimand means, why the answers differ, and which claims remain unsupported.

SYNTHESIZE Explain the two Prestige answers without saying that one controls away bias.

Research check: Conditioning changes the comparison. Whether it improves a causal answer requires a separate design and substantive argument.

Prestige evaluation: alignment, evidence, and inference

Lens	What it reveals
Question alignment	Both questions name occupations and linear association; RQ2 explicitly changes the comparison through conditioning.
Evidence integrity	Measures are historical occupational aggregates; missing type for four occupations narrows the empirical basis to the common 98 and none of the measures describes individual workers.
Justification adequacy	Linear models support statistical summaries, but observational aggregation does not support an individual or causal interpretation.
Inferential adequacy	The common 98 separate coefficient change from sample change; additive form, classical error conditions, overlap, and influence diagnostics remain consequential.

EVALUATE Which of these four lenses most limits an individual-level interpretation?

Research check: The occupation is the unit, so more precise regression cannot create person-level evidence.

Prestige evaluation: transition and claim calibration

Lens	What it reveals
Transition fidelity	Each substantive interpretation retains its own slope, interval, unit, assumptions, and exploratory status.
Claim calibration	RQ1 supports a bounded occupation-level association; RQ2 remains inconclusive and model-dependent.
Cross-question coherence	The synthesis explains why the answers differ instead of choosing one coefficient as the universal result.

Prompted revision: report both estimands and both answer outcomes. Any individual-level or causal question would require different evidence and support, not more confident wording.

EVALUATE Which lens most constrains each Prestige answer, and why are the constraints different?

Research check: Shared evidence does not make the two estimands or claims interchangeable.

What the real application integrates

- ▶ **Argument:** each question has a separate statistical target and bounded substantive answer, even though much evidence is shared.
- ▶ **Evaluation:** aggregation, missing classification, model form, influence, and exploratory status narrow both claims in different ways.
- ▶ **Evidence and design:** occupations are the units, the measures are historical aggregates, and the observational design supports association but not an individual return or causal effect.

The available record supports Argument and Evaluation but does not establish the original researchers' planning and collection sequence well enough to reconstruct their Workflow.

The aim is not to choose the “real coefficient.” It is to explain what each question can establish and why the two answers can coexist.

SYNTHESIZE Explain what the example demonstrates without reconstructing an undocumented workflow.

Research check: The available record supports argument and evaluation, not claims about how the original study was planned.

Repairing an overclaim requires both arguments

Overclaim

An extra year of education causes occupational income to rise by CAD 883.

Statistical repair

Under the marginal occupation-level model, the estimated income slope is CAD 883 per additional average education year, 95% CI [CAD 628, CAD 1,138].

Substantive repair

Among the represented occupations, one additional average education year was associated with CAD 883 higher average income, 95% CI [CAD 628, CAD 1,138], in 1971 Canadian dollars; aggregation, observational design, and alternative explanations preclude an individual or causal interpretation.

The repair retains the numerical evidence while changing the unit, verb, and scope to match the question and support.

REPAIR Replace the causal verb, restore the occupation unit, and retain the interval and scope.

Research check: A repair changes the claim, not the observed estimate.

A compact explanation follows the support structure

When reading or writing empirical research, ask:

1. What quantitative question is being answered?
2. What does the empirical evidence represent?
3. What statistical and substantive support connects that evidence to the intended targets?
4. What was estimated or predicted, with what uncertainty and checking?
5. What does the result mean in this context?
6. What bounded statistical and substantive claims follow?
7. Is the answer outcome bounded or inconclusive, and is its analytic status confirmatory or exploratory?

This sequence is not a rigid paragraph template. It is a way to ensure that procedural detail contributes to a traceable answer.

WRITE Use the seven prompts to draft one traceable result paragraph.

Research check: Every procedural detail should support the question, inference, or bounded answer.

A study-level synthesis comes after question-level answers

For each research question, record:

- ▶ its estimand or predictive target;
- ▶ its statistical and substantive arguments;
- ▶ its answer outcome and analytic status; and
- ▶ its unresolved reservations.

Only then synthesize across questions. Shared evidence and support should be identified once and referenced clearly, while different estimands and claims remain distinct.

A study is therefore a collection of linked arguments followed by a coherent synthesis—not one broad argument followed by unrelated coefficients.

SYNTHESIZE Combine question-specific claims only after each has its own target and reservations.

Research check: A study-level synthesis must not blur distinct estimands.

Regression is an inferential grammar, not a source of meaning

Regression expresses a conditional mean or another outcome feature as a function of predictors. It can represent:

- ▶ group comparisons and ANOVA;
- ▶ adjustment for multiple predictors;
- ▶ interaction and moderation;
- ▶ binary, ordinal, nominal, count, and rate outcomes; and
- ▶ predictive models with regularization.

Regression helps specify estimands, calculate estimates, quantify uncertainty, and check models. It does not by itself provide representative sampling, valid measurement, causal identification, or substantive importance.

CONNECT Identify which part of the argument regression can develop and which it cannot.

Research check: Regression is a language for comparisons and prediction, not a source of measurement or causal support.

Later chapters develop different parts of the arguments

- ▶ **Chapters 2–4:** statistical concepts, basic inference, categorical association, and ANOVA.
- ▶ **Chapters 5–6:** R foundations, evidence preparation, statistical computing, visualization, and basic inference.
- ▶ **Chapters 7–9:** simple and multiple regression, then interaction, moderation, moderated moderation, and conditional-process concepts.
- ▶ **Chapters 10–12:** binary, nominal, ordinal, count, and rate regression.
- ▶ **Chapter 13:** prediction, cross-validation, and regularization.
- ▶ **Chapter 14:** Simpson's paradox, regression to the mean, other pitfalls, and credible claims.

Every method enters through a quantitative question. The recurring task is to connect evidence and support to statistical inference, then develop a bounded substantive answer when the question requires one.

MAP Choose one later chapter and identify which component or evaluation lens it deepens.

Research check: New methods extend the same question-centred reasoning.

The framework closes where the research question began

A defensible quantitative study can be read in one sentence:

A clearly framed question is addressed through empirical evidence, appropriate statistical and substantive support, calibrated inference, and a bounded answer whose uncertainty and reservations remain visible.

The framework offers three complementary views:

- ▶ **Argument** explains what must justify the answer.
- ▶ **Workflow** explains how the reasoning is constructed and revised.
- ▶ **Evaluation** explains how its quality is examined and calibrated.

The method matters because it helps answer the question—not because operating the method is the purpose of research.

RESTATE Answer one quantitative question while preserving evidence, support, uncertainty, and reservations.

Research check: The method matters only through the defensibility of the answer.

Reading bridge

- ▶ Purdue OWL: Toulmin argument — accessible orientation to grounds, warrant, backing, qualifier, claim, and rebuttal.
- ▶ ModernDive: Simple and multiple regression — public regression examples connecting questions, models, and interpretations.
- ▶ OpenIntro Statistics — public foundations for study design, sampling, estimation, and inference.
- ▶ R for Data Science (2e) — public reference for reproducible data work and visualization.

Read each source through all three perspectives: What question is being answered? What support structure is offered? How was it constructed? Which evaluation lens most constrains the answer?

CONNECT Choose one reading and identify the question, STAT-C, and SUB-C.

Research check: A source may teach a method well while leaving parts of its substantive argument implicit; the framework makes those omissions discussable.

Toulmin's roles and the framework should remain distinguishable

Toulmin asks how grounds support a claim through a warrant, with backing, qualifier, and rebuttal. The present framework asks how a quantitative research question receives statistical and substantive answers through eight question-relative components.

Useful correspondences exist:

- ▶ empirical evidence often supplies grounds;
- ▶ statistical and substantive justification perform warrant-like and backing-like work;
- ▶ uncertainty shapes the force of a claim; and
- ▶ reservations delimit the answer.

The correspondence is intentionally approximate. The framework organizes two linked empirical arguments and three research perspectives rather than renaming Toulmin's six roles.

COMPARE Match one framework role to a Toulmin function, then name why the match is incomplete.

Research check: The framework coordinates two arguments rather than renaming Toulmin's six roles.

Measurement questions can make validity the object of inquiry

When the question asks whether a scale supports a stated interpretation, indicator responses and administration information form empirical evidence; factor structure, reliability, invariance, and external relations contribute to statistical and substantive inference; and a bounded validity answer is SUB-C. Component roles are question-relative. Full latent-variable modelling belongs to Part Two.

REVERSE When validity is the question, identify what becomes the final substantive claim.

Research check: Component roles depend on the question being answered.

A quasi-experiment needs design-specific support

A policy threshold, interrupted time series, or externally imposed timing can create informative contrasts without random assignment. Credibility depends on the design:

- ▶ Can units manipulate threshold placement?
- ▶ Would the outcome trend have continued without the intervention?
- ▶ Did another event occur at the same time?
- ▶ Is the comparison local or broad?

A regression estimate supplies **STAT-I**. The quasi-experimental logic and checks must support **STAT-J** and **SUB-J** before any causal **SUB-C** is defensible.

JUSTIFY Name the design-specific assumption that makes the quasi-experimental contrast informative.

Research check: A regression estimate cannot replace identification logic.

Sensitivity analysis asks how claims change

A sensitivity analysis varies a consequential assumption or analytic decision within a substantively defensible range. It can examine:

- ▶ missingness assumptions;
- ▶ influential observations;
- ▶ functional form;
- ▶ measurement error;
- ▶ plausible unmeasured confounding; or
- ▶ alternative but defensible outcome definitions.

The result belongs in **STAT-I** and then shapes **STAT-C**, **SUB-I**, and **SUB-C**.

Repeating many analyses without explaining what uncertainty each one addresses is not a credibility argument.

VARY State one plausible assumption change and predict which claims it could alter.

Research check: Sensitivity analysis is useful only when its changed condition has a clear inferential role.

Final optional practice: compare two claims

Claim A

The intervention worked because the coefficient was statistically significant.

Claim B

Under the prespecified model, assignment was associated with a 3.1-point increase, 95% CI [0.8, 5.4]. Randomization supports an assignment-based causal interpretation for the consenting students in this implementation; the magnitude relative to the 2-point benchmark and wider generalization remain uncertain.

Use the seven lenses to explain why Claim B is stronger. Then identify one piece of evidence or support that would still be needed before extending the claim to another institution.

COMPARE Use the seven lenses to explain why Claim B is stronger and still bounded.

Research check: More detail helps only when it preserves evidence, design, uncertainty, and scope.

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